Statistical Estimation of Canteen Arrival Overload Probability Using Poisson Distribution

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***Abstract*—University canteens see sharp, short-lived rises in arrivals during the lunch peak, and the practical question for students and managers is how likely a given time slot is to exceed what the serving counters can handle. This paper models student arrivals in each 15-minute slot as a Poisson count, estimates a separate rate λ_t for every slot with an exact confidence interval, tests the Poisson assumption slot by slot, and converts the tail probability $P(X_t > C)$ into low, medium and high overload-risk labels. The analysis is then extended from description to prediction: a multiple linear regression on log counts gives day-specific rates for each weekday and for exam weeks; a Poisson–Gamma mixture accounts for day-to-day variation; Monte Carlo simulation gives the probability that each slot is the day's peak; and Bayes' theorem with a Gamma prior updates the forecast from the first counts of the day. Every forecast is checked on days the model has not seen with a rolling one-day-ahead backtest. On a reproducible synthetic semester of 90 working days, the riskiest slot is 13:15–13:30 ($P = 73.9\%$ at $C = 30$); on a regular Monday the peak falls in 13:00–13:30 with probability 75%; forecast intervals cover 93.7% of unseen counts and the busiest slot is named correctly 40% of the time against 13% by chance, while the gain in point accuracy over a per-slot average is not statistically significant. All methods use standard probability and statistics, and the implementation is released as an interactive dashboard.**

***Index Terms*—Poisson distribution, arrival process, overload probability, peak-time forecasting, Bayesian updating, negative binomial, hypothesis testing, university canteen.**

I. Introduction

University canteens experience changing arrival intensity during lunch. When more students arrive in a short interval than the counters can serve, queues form, the dining area becomes crowded and popular dishes run out. Decisions about when to eat, how many counters to open and when to stagger breaks are usually taken by intuition rather than from measured arrival behaviour.

Existing canteen studies treat the canteen as a complete system. Tang *et al.* [1] collected camera data at a university canteen and built a pedestrian-flow model of the whole meal process, and a cellular-automaton model followed [2]. Queueing studies set the number of serving windows from assumed Poisson arrivals [3]–[5]. These models are valuable but heavy, and they leave a simple question unanswered: *what is the probability that arrivals exceed a chosen level during a particular time slot, and when will today's peak occur?*

This paper answers that question with standard probability and statistics. Its contributions are:

1) A per-slot Poisson model of canteen arrivals with exact confidence intervals, slot-by-slot goodness-of-fit and dispersion tests, and an overload probability $P(X_t > C)$ mapped to risk labels.
2) A day-specific forecast of slot rates by multiple linear regression on weekday and exam-week indicators, with a Poisson–Gamma correction for day-to-day variation.
3) The probability that each slot is the day's peak, obtained by simulation, and a Bayesian update of the rest-of-day forecast from the counts observed so far.
4) A validation protocol that forecasts each day from earlier days only and tests the forecasts with proportion, paired, sign, Kolmogorov–Smirnov and chi-square tests, and the Brier score.

Field data collection is in progress, so all results here are computed on a clearly labelled synthetic semester (Section III); the methods apply unchanged to field counts.

The rest of the paper is organised as follows. Section II reviews related work and the gaps it leaves. Section III describes the data and Section IV the methodology. Section V presents results, Section VI discusses them and their limitations, and Section VII concludes.

II. Related Work

A. Canteen flow and queueing

Tang *et al.* [1] counted arrivals and departures in 5-minute segments at a Beihang University canteen between 10:30 and 13:30 and calibrated a model of window choice, movement, congestion delay and meal duration. They note that arrivals

follow a Poisson distribution within short periods with a rate that changes every few minutes, and that days differ with the class timetable; the arrival rate, however, is an input to their simulation rather than a quantity that is tested or predicted. Queueing studies set windows, seats and opening times with M/M/n/m and M/M/C models [3], [4]; Jiao *et al.* [5] simulate one constant arrival rate of 19.3 students per minute and find seven windows optimal. In all of these the Poisson arrival assumption is an input, not a finding.

B. Testing Poisson arrivals

Brown *et al.* [6] established a statistical treatment of service arrivals and a test that a point process is Poisson. Kim and Whitt [7] show that nonhomogeneous Poisson models fit call-centre and hospital arrivals only after careful analysis: long subintervals and pooled days create overdispersion. Green *et al.* [8] show that time-varying demand needs short, piecewise-constant rates, and Leemis [9] estimates such rates from repeated days. Cameron and Trivedi [10] give tests for overdispersion in Poisson models.

C. Forecasting arrival rates

Weinberg *et al.* [11] forecast within-day call-centre rates as a daily level times a weekday shape and update the forecast during the day; Shen and Huang [12] forecast quarter-hour Poisson rate profiles and report that intraday updating cut staffing error by 14.5%. Aldor-Noiman *et al.* [13] add special-event indicators and compare against an "industry model" that averages past slots, which is the per-slot average used by descriptive canteen analyses. Matteson *et al.* [14] stress that small hourly counts should be modelled as Poisson directly rather than through normal approximations. Ibrahim *et al.* [15] survey the field: arrivals are overdispersed and correlated within a day, which a Gamma "busyness factor" captures, and simple fixed-effect models are hard to beat.

D. Campus dining and forecast verification

Shah *et al.* [16] forecast daily customer totals in a university self-service restaurant with machine learning, although their own data show strong hourly peaks; Variskhanov and Arreeras [17] issue a day-level "early peak" badge from Wi-Fi telemetry. For probability forecasts of a yes/no event, the Brier score [18] and a skill score against a reference forecast [19] are the standard checks.

E. Research gaps

Table I summarises what the reviewed work leaves open. Four gaps motivate this paper: (G1) no per-slot overload probability for a canteen; (G2) the Poisson assumption is built in rather than tested; (G3) no prediction of a specific day's rates or of the peak time, with or without the day's early counts; and (G4) no test of canteen forecasts on unseen days. These are gaps of focus and application, not a claim that Poisson modelling of arrivals is new.

TABLE I
REPRESENTATIVE STUDIES AND WHAT THEY LEAVE OPEN

Study	Approach	Left open for this work
Tang *et al.* [1]	Pedestrian-flow model of a canteen	Arrival rate is an input; no slot overload probability
Jiao *et al.* [5]	M/M/C simulation, one rate	Poisson assumed; rate constant all lunch
Kim & Whitt [7]	Tests of Poisson arrivals	Call centres and hospitals, not canteens
Weinberg *et al.* [11], Shen & Huang [12]	Rate forecasting, intraday update	Large-count approximations; staffing, not overload
Shah *et al.* [16]	ML forecast of campus dining	Daily totals only; no probabilities

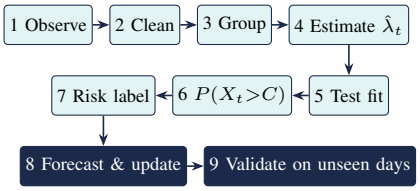


Fig. 1. Statistical framework. Light boxes: estimation and checking; dark boxes: prediction and validation.

III. Data

Collection protocol. Arrivals are counted at the canteen entrance, manually or from permitted video, in fixed 15-minute slots from 12:30 to 14:30 on m working days, following the observation style of [1]. Each record holds the date, the slot, the arrival count and, optionally, the day type (regular or exam week).

Prototype data. Until field collection is complete, all results use a synthetic semester generated with a fixed random seed: 90 working days (18 weeks) $\times$ 8 slots. Expected arrivals follow a background rate of 14 per slot plus a Gaussian lunch rush centred at 13:15; each day receives a Gamma busyness factor with mean 1 and 5% coefficient of variation and a weekday effect (Friday 6% quieter); weeks 9 and 18 are exam weeks with demand reduced by 15%; counts are drawn from the resulting Poisson distributions and 0.5% are blanked to exercise the cleaning step. The analysis never sees these parameters and re-estimates everything from the counts.

Cleaning. Missing, negative, fractional, unreadable and duplicate counts are dropped rather than imputed, because imputing the mean would shrink the variance and bias the dispersion test. Three of 720 records were missing, leaving 717 slot observations.

IV. Methodology

Fig. 1 shows the nine-step framework. Steps 1–7 estimate and check overload risk; steps 8–9 forecast and validate.

A. Poisson model and estimation

Let X_t be the number of students arriving in slot t. The model assumes

$$X_t \sim \text{Poisson}(\lambda_t), \qquad P(X_t = x) = \frac{e^{-\lambda_t}\lambda_t^x}{x!}, \qquad (1)$$

$x = 0, 1, 2, \ldots$, with $E[X_t] = \mathrm{Var}(X_t) = \lambda_t$. For slot t observed on m days with counts $x_1, \ldots, x_m$, the maximum-likelihood estimate is the sample mean $\hat{\lambda}_t = \sum x_i / m$. Because $S = \sum x_i \sim \mathrm{Poisson}(m\lambda_t)$, the exact (Garwood) interval is [20]

$$\left[\frac{\chi^2_{\alpha/2}(2S)}{2m}, \; \frac{\chi^2_{1-\alpha/2}(2S+2)}{2m} \right]. \tag{2}$$

B. Checking the Poisson assumption

Two tests are applied to every slot at $\alpha = 0.05$. The index-of-dispersion test uses $D = (m-1)s^2/\bar{x} \sim \chi^2_{m-1}$ under H_0, two-sided. The chi-square goodness-of-fit test compares observed and expected frequencies, $\chi^2 = \sum_i (O_i - E_i)^2 / E_i$ with $k - 2$ degrees of freedom, after merging adjacent counts until every expected frequency is at least 5 [21]. Because eight slots are tested, the number of rejections is compared with a $\mathrm{Binomial}(8, \alpha)$ distribution.

C. Overload probability and risk

With capacity C per slot (threshold $K = C + 1$),

$$P(X_t > C) = 1 - \sum_{x=0}^{C} \frac{e^{-\hat{\lambda}_t} \hat{\lambda}_t^x}{x!}. \tag{3}$$

A slot is labelled low risk below 20%, high risk above 50% and medium otherwise; these thresholds are project decision rules. The probability is also evaluated at both ends of the confidence interval for λ_t, and the smallest capacity keeping it below a target is reported for planning. To check that the quietest slot is genuinely quieter than the runner-up, the exact conditional test $S_1 \mid (S_1 + S_2) \sim \mathrm{Binomial}(S_1 + S_2, m_1/(m_1 + m_2))$ is used [22].

D. Day-specific rates by regression

To predict a particular day, the slot rate is allowed to depend on the weekday d and the day type e. Multiple linear regression with dummy variables is fitted by least squares to log counts:

$$\ln(x_{dt} + 0.5) = \beta_0 + \alpha_t + \gamma_d + \delta_e + \varepsilon_{dt}, \tag{4}$$

with 12:30, Monday and regular weeks as reference levels. Each coefficient is tested with $t_j = \hat{\beta}_j / \mathrm{SE}(\hat{\beta}_j) \sim t_{n-p}$, and fitted values are returned to the count scale with Duan's smearing factor $\overline{e^{\hat{\varepsilon}}}$ [23]. Weekday effects on daily totals are also tested with one-way ANOVA and the Kruskal–Wallis test, and exam weeks with Welch's t-test and the Mann–Whitney U test [24].

E. Day-to-day variation: Poisson–Gamma model

Whole days can be busier or quieter than predicted [15]. Let each day carry a busyness factor $W \sim \mathrm{Gamma}(1/v, 1/v)$ with mean 1 and variance v. By the theorem of total probability,

$$P(X_t = x) = \int_0^\infty \frac{e^{-w\lambda_t} (w\lambda_t)^x}{x!} \, g(w) \, dw, \tag{5}$$

a negative binomial with $E[X_t] = \lambda_t$ and $\mathrm{Var}(X_t) = \lambda_t + v\lambda_t^2$. The Pearson statistic $\chi^2_P = \sum (x - \hat{\lambda})^2 / \hat{\lambda}$ follows χ^2_{n-p} under the plain Poisson model, and its excess gives the moment estimate $\hat{v} = (\chi^2_P - (n-p)) / \sum \hat{\lambda}$ [10]. With $v = 0$ the model reduces to the Poisson model above.

F. Peak-time probability

For a chosen day, $N = 20{,}000$ lunch periods are simulated: draw W, then $X_t \sim \mathrm{Poisson}(W\hat{\lambda}_t)$ independently for each slot. The share of simulated days on which slot t has the largest count (ties broken at random) estimates $P(\text{slot } t \text{ is the peak})$; the same draws give $P(\text{at least one slot exceeds } C)$.

G. Same-day Bayesian update

After the first k slots of today have been counted, Bayes' theorem with the Gamma prior $W \sim \mathrm{Gamma}(a, a)$, $a = 1/v$, gives the conjugate posterior

$$W \mid x_1, \ldots, x_k \sim \mathrm{Gamma}\Big(a + \sum_{t \le k} x_t, \; a + \sum_{t \le k} \hat{\lambda}_t\Big), \tag{6}$$

and each later slot then has a negative binomial predictive distribution, which yields the updated $P(X_t > C)$.

H. Validation on unseen days

Each day from day 31 onward is forecast from all earlier days only and compared with what happened. The model is compared with the per-slot average used in descriptive analyses (the "industry model" [13]). The checks are: coverage of 95% prediction intervals (z-test of a proportion); whether the predicted busiest slot is correct more often than chance (one-sided z-test); daily mean absolute error against the baseline (paired t-test and sign test); calibration of the full forecast distribution through randomised probability integral transform (PIT) values, $u = F(x-1) + V[F(x) - F(x-1)]$ with $V \sim U(0,1)$, tested for uniformity by Kolmogorov–Smirnov and chi-square tests; and the Brier score $B = \frac{1}{N}\sum (p_i - o_i)^2$ of the overload forecasts with skill $1 - B_{\text{model}}/B_{\text{baseline}}$ [18], [19]. Spearman rank correlation between consecutive slots and a Wald–Wolfowitz runs test on daily residual signs check within-day and day-to-day dependence.

I. Implementation

The methods are implemented in Python with NumPy, pandas and SciPy, covered by 83 automated tests, and presented in an interactive Streamlit dashboard whose home page answers "when should I go?" in plain language. All figures and tables in this paper are generated from the same code.

V. Results

All results use the 90-day synthetic semester, capacity $C = 30$ per slot and $\alpha = 0.05$.

A. Arrival rates and model fit

Arrival rates rise from 24.5 at 12:30 to a peak of 34.4 at 13:15–13:30 and fall to 16.3 at 14:15–14:30 (Fig. 2, Table II). The confidence intervals of the three central slots lie partly above the capacity line, so an average day already exceeds capacity there. The chi-square test rejects the Poisson model in none of the eight slots, against 0.4 rejections expected by

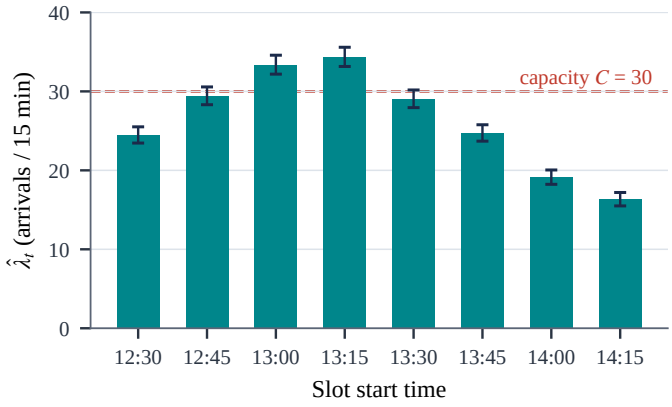


Fig. 2. Estimated arrival rate per slot with exact 95% confidence intervals and the capacity $C = 30$.

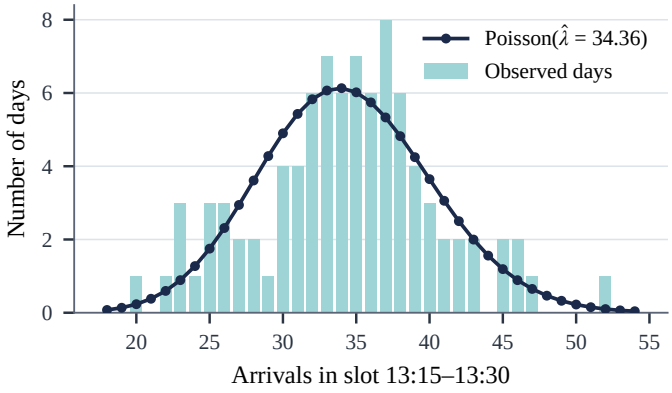


Fig. 3. Observed distribution of daily counts in the peak slot against the fitted Poisson model.

chance; for the peak slot, $\chi^2 = 8.14$ on 11 degrees of freedom, $p = 0.701$ (Fig. 3). The quietest slot, 14:15–14:30 ($\hat{\lambda} = 16.3$), is significantly quieter than the runner-up 14:00–14:15 ($\hat{\lambda} = 19.1$; exact test $p = 5.2 \times 10^{-6}$).

B. Overload risk

Two slots, 13:00–13:15 and 13:15–13:30, are high risk ($P = 68.2\%$ and 73.9%), two are medium and four are low (Fig. 4). Keeping $P(X_t > C)$ below 20% in every slot would need a capacity of 39 per slot at the peak, against 30 at present. The Pearson test detects day-to-day overdispersion once weekday and exam effects are removed ($\chi^2_P = 812.4$ on 704 df, $p = 0.003$; $\hat{v} = 0.0057$, a coefficient of variation of 7.6%). The Poisson–Gamma correction moves the probabilities only slightly, lowering them near the peak (71.8% against 73.9%) and raising them in the quieter tails.

C. Day-specific forecast and peak time

The regression explains 57.9% of the variance of log counts (adjusted $R^2 = 0.572$). Relative to a regular Monday, Friday multiplies expected arrivals by 0.906 ($t = -3.83$, $p < 0.001$), Thursday by 0.946 ($p = 0.032$) and exam weeks by 0.887 ($t = -4.62$, $p < 0.001$); Tuesday and Wednesday are not significant (Table III). On daily totals the weekday effect is only

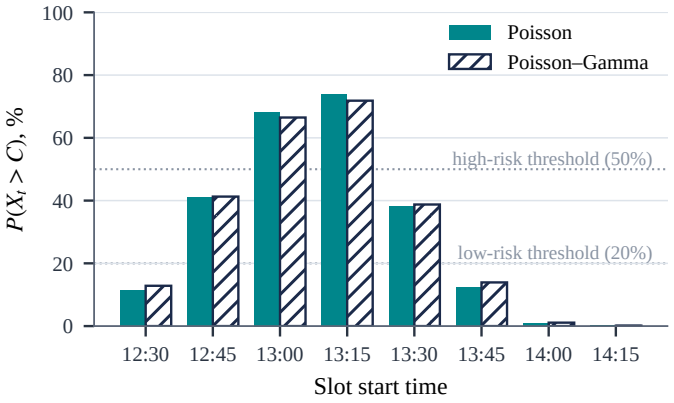


Fig. 4. Overload probability per slot under the Poisson and Poisson–Gamma models, with the risk thresholds.

borderline (ANOVA $p = 0.091$, Kruskal–Wallis $p = 0.089$), whereas exam days are clearly quieter (191.9 against 212.4 arrivals per lunch; Welch $t = -4.37$, $p < 0.001$; Mann–Whitney $p = 0.005$).

Fig. 5(a) shows the resulting forecasts. On a regular Monday the probability that at least one slot is overloaded is 97.9%, with 3.0 overloaded slots expected; on a regular Friday these fall to 90.0% and 2.0, and on an exam-week Monday to 87.1% and 1.8. The peak is most likely at 13:15–13:30 (42%), with 13:00–13:15 close behind (33%), so the peak falls in 13:00–13:30 with probability 75% (Fig. 5(b)). Neighbouring slots have similar rates, so the peak time is genuinely uncertain on any single day even when the rates are known.

D. Same-day update

Suppose the first two slots of a regular Monday bring 30 and 36 students, about 15% more than forecast. The posterior mean of the busyness factor becomes 1.041 (standard deviation 0.067), and the overload probability at 13:15–13:30 rises from 80.7% to 86.6%, and at 13:30–13:45 from 49.4% to 57.8% (Fig. 6). Because the estimated day-to-day variation is small, the prior is strong and a busy morning moves the forecast moderately; larger variation in field data would move it further.

E. Validation on unseen days

Sixty days (477 slot forecasts) were forecast one day ahead. Table IV summarises the tests and Fig. 7 the calibration. The 95% prediction intervals covered 93.7% of actual counts, consistent with the nominal level ($p = 0.196$). The predicted busiest slot was correct on 40.0% of days against 13.4% by chance ($z = 6.04$, $p < 0.001$). Predicted and actual counts correlate with $r = 0.727$. The daily mean absolute error was 4.57 for the model against 4.62 for the per-slot average, a difference that is not significant (paired t, $p = 0.290$; sign test 35 of 60 days, $p = 0.123$); the gain is concentrated on exam days (4.53 against 4.78). The Brier scores of the overload forecasts are almost equal (0.1394 against 0.1393). Updating after the first two slots improves the Brier score of the later slots by 4.0% (0.1191 against 0.1240). The PIT histogram is flat by the

TABLE II
PER-SLOT ESTIMATES, MODEL CHECKS AND OVERLOAD PROBABILITIES ($C = 30$, 90 DAYS)

Slot	$\hat{\lambda}_t$	95% CI	Var/Mean	χ^2 GOF p	$P(X_t > C)$ Poisson, %	$P(X_t > C)$ Poisson–Gamma, %	Risk
12:30–12:45	24.47	[23.46, 25.51]	1.09	0.450	11.38	12.84	Low
12:45–13:00	29.42	[28.31, 30.56]	1.34	0.552	40.98	41.27	Med.
13:00–13:15	33.37	[32.18, 34.58]	1.26	0.309	68.23	66.48	High
13:15–13:30	34.36	[33.16, 35.59]	1.10	0.701	73.94	71.84	High
13:30–13:45	29.04	[27.94, 30.18]	1.03	0.218	38.26	38.75	Med.
13:45–14:00	24.72	[23.70, 25.77]	1.25	0.310	12.43	13.92	Low
14:00–14:15	19.12	[18.23, 20.05]	1.34	0.148	0.76	1.09	Low
14:15–14:30	16.33	[15.50, 17.20]	1.24	0.257	0.08	0.14	Low

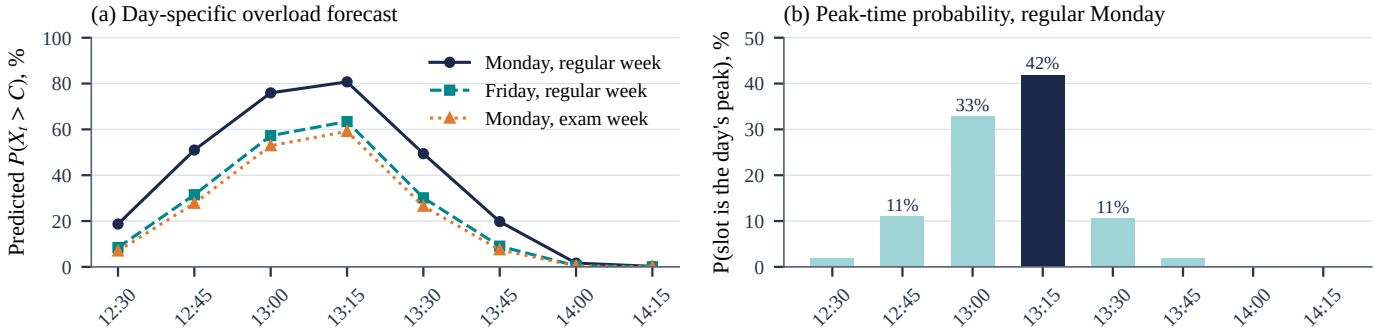


Fig. 5. (a) Predicted overload probability for three kinds of day. (b) Probability that each slot is the busiest of the day on a regular Monday, from 20,000 simulated lunch periods.

TABLE III
REGRESSION OF $\ln(x + 0.5)$ ON SLOT, WEEKDAY AND DAY TYPE

Term	Estimate	SE	t	p	$e^{\hat{\beta}}$
Intercept†	3.2441	0.0282	114.90	<0.001	25.638
Slot 12:45	0.1792	0.0324	5.53	<0.001	1.196
Slot 13:00	0.3081	0.0324	9.51	<0.001	1.361
Slot 13:15	0.3386	0.0324	10.45	<0.001	1.403
Slot 13:30	0.1716	0.0324	5.30	<0.001	1.187
Slot 13:45	0.0077	0.0325	0.24	0.812	1.008
Slot 14:00	−0.2570	0.0324	−7.93	<0.001	0.773
Slot 14:15	−0.4100	0.0326	−12.59	<0.001	0.664
Tuesday	−0.0179	0.0257	−0.69	0.487	0.982
Wednesday	0.0003	0.0257	0.01	0.991	1.000
Thursday	−0.0552	0.0257	−2.15	0.032	0.946
Friday	−0.0985	0.0257	−3.83	<0.001	0.906
Exam day	−0.1198	0.0259	−4.62	<0.001	0.887

†Reference levels: slot 12:30, Monday, regular week. Slots are labelled by start time.

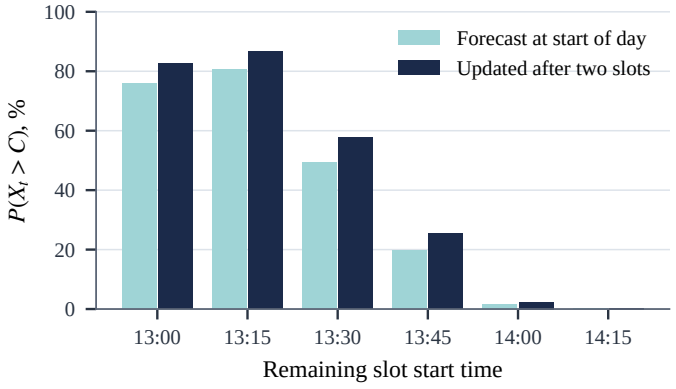


Fig. 6. Overload probability of the remaining slots before and after observing the first two slots of the day.

chi-square test ($p = 0.062$) but the Kolmogorov–Smirnov test rejects uniformity ($p = 0.028$): an excess of PIT values near 1 shows that the highest counts are somewhat under-forecast. Across eight independently simulated semesters the model's error was below the baseline's in all eight, significantly in four.

Within-day carry-over is weak: Spearman correlations of observed-to-predicted ratios between consecutive slots are positive and significant for two of seven pairs ($\rho = 0.235$, $p = 0.026$ and $\rho = 0.222$, $p = 0.038$). The runs test finds no clustering of busy days (42 runs against 45.8 expected, $p = 0.42$).

VI. DISCUSSION

The per-slot Poisson model fits every slot and turns arrival counts into a direct answer: in this semester the 15 minutes from 13:15 are overloaded on roughly three days in four, and 14:15 is reliably the quietest time. Testing the assumption slot by slot, rather than building it in as queueing studies do [3]–[5], is what makes the probabilities defensible; the pooled Pearson test then shows that the per-slot tests alone can miss day-to-day variation, which the Poisson–Gamma model absorbs.

The forecasting results are mixed and are reported as such. The model's intervals are well calibrated, it names the peak slot three times more often than chance, it captures the exam-

TABLE IV
HYPOTHESIS TESTS OF THE ONE-DAY-AHEAD FORECASTS (60 UNSEEN DAYS, 477 SLOT FORECASTS, $\alpha = 0.05$)

Check	Test	Statistic	p	Result	Supports the model
95% interval coverage	z-test, one proportion	$z = -1.29$	0.196	coverage 93.7%	Yes
Peak slot named correctly	z-test, one proportion	$z = 6.04$	<0.001	hit rate 40.0% vs chance 13.4%	Yes
Lower daily MAE than baseline	Paired t-test	$t = 0.56$	0.290	MAE 4.57 vs 4.62	No
More winning days than baseline	Sign test	35 wins, 25 losses	0.123	35/60 days better	No
PIT uniform	Kolmogorov–Smirnov	$D = 0.067$	0.028	not uniform	No
PIT histogram flat	χ^2 goodness of fit	$\chi^2 = 16.23$, df = 9	0.062	flat	Yes

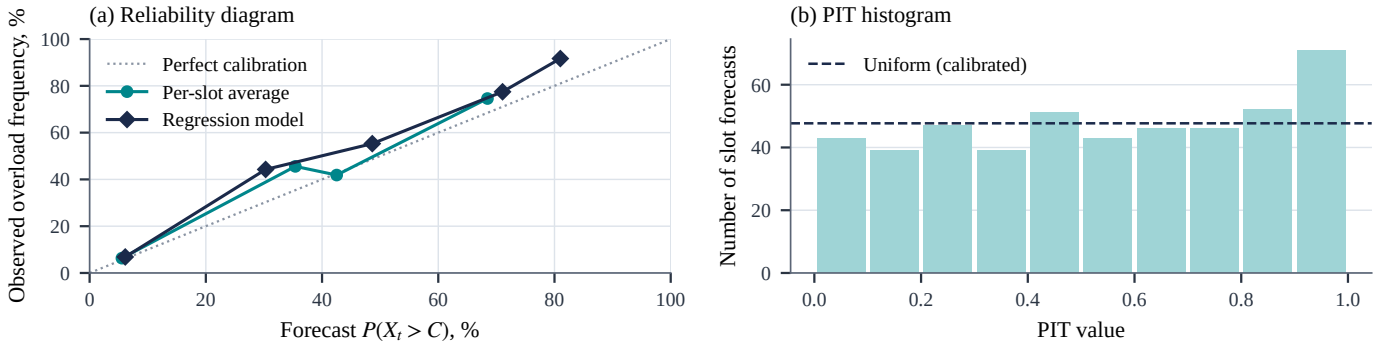


Fig. 7. Calibration on unseen days. (a) Forecast overload probability against observed overload frequency. (b) Randomised PIT values of the count forecasts; a flat histogram indicates a calibrated forecast distribution.

week drop that a per-slot average misses, and the same-day update measurably improves the afternoon forecast. Its point accuracy, however, is not significantly better than the per-slot average on this semester, in line with the observation that simple fixed-effect models are hard to beat when day effects are mild [15]. The value of the forecasting layer therefore lies in what the average cannot give: day-type forecasts, peak-time probabilities and updating during the day.

Limitations. (i) The data are synthetic; conclusions about any real canteen await field counts. (ii) The model estimates arrival risk, not queue length, which also depends on service rates and counters. (iii) The rate is assumed constant within a slot and arrivals independent, whereas groups of friends arrive together. (iv) The regression scales a whole day up or down and cannot move the peak to another slot on special days without slot-by-day-type interactions. (v) Least squares on log counts stands in for Poisson regression and slightly under-forecasts the highest counts. (vi) Capacity and risk thresholds are project decisions that must be justified for a specific canteen.

VII. CONCLUSION AND FUTURE WORK

This paper estimated the probability that canteen arrivals exceed capacity in each 15-minute slot using a Poisson model checked slot by slot, and extended it to predict a specific day's risk and peak time. The methods identified a high-risk window at 13:00–13:30, a significantly quieter time at 14:15, a 75% probability that a regular Monday's peak falls in 13:00–13:30, and a measurable benefit from updating the forecast with the day's first counts, while showing honestly that the gain in point accuracy over a per-slot average is not significant on the prototype data.

Future work will replace the synthetic semester with entry counts collected at the canteen, add service times to estimate queue length through an M/M/c model, model group arrivals with compound Poisson distributions, estimate a smooth rate $\lambda(t)$ [9], and run the same-day update automatically from turnstile or camera counts.

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